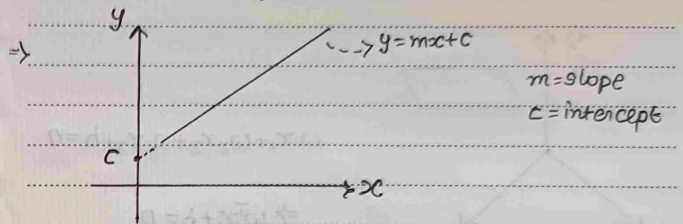


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### Equation of a Line, Plane, Hyperplane



Also,  $ax + by + c = 0 \rightarrow$  another eq. of line.

How?

$$ax + by + c = 0 \Rightarrow y = \underbrace{\left(-\frac{a}{b}\right)}_m x - \underbrace{\frac{c}{b}}_c$$

Also, for  $x_1, x_2$  axis:

$$w_1 x_1 + w_2 x_2 + b = 0 \Rightarrow w^T x + b = 0$$

intercept

coefficients

another eq. of line



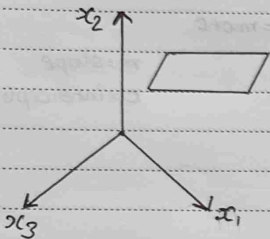
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⇒ For 3D plane:



$$\omega_1 x_1 + \omega_2 x_2 + \omega_3 x_3 + b = 0$$

$$\Rightarrow \omega^T x + b = 0$$

where

$$\omega = \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}, x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

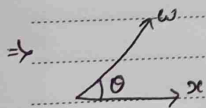
⇒ For  $\pi$ -D plane:

$$\omega_1 x_1 + \omega_2 x_2 + \dots + \omega_n x_n + b = 0$$

$$\Rightarrow \omega^T x + b = 0$$

⇒ Eq. of st. line passing through origin:

$$\omega^T x = 0$$



$$\omega \cdot x = \omega^T x = |\omega| |x| \cos \theta = 0$$

$$\Rightarrow \theta = 90^\circ$$

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Date .....

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